```python
#1.Print all even numbers from 1 to 50 (no arguments)
def print_even():
    for i in range(1, 51):
        if i % 2 == 0:
            print(i)

print_even()
```

```python
#2. Function to print sum of two numbers
def add(a, b):
    print("Sum =", a + b)

add(10, 20)
```

```python
#3. Return multiple values (sum, average, min, max) from a list
def list_details(lst):
    total = sum(lst)
    avg = total / len(lst)
    return total, avg, min(lst), max(lst)

numbers = [10, 20, 30, 40]
```

```python
s, a, mn, mx = list_details(numbers)

print("Sum:", s, "Average:", a, "Min:", mn, "Max:", mx)



#4. Largest of three numbers using return

def largest(a, b, c):

    return max(a, b, c)


print("Largest:", largest(10, 25, 15))



#5. Positional and keyword arguments together

def student_details(name, roll, course="Python"):

    print("Name:", name)

    print("Roll No:", roll)

    print("Course:", course)


student_details("Arun", 101, course="AI")



#6. Area of a rectangle (positional arguments)

def area_rectangle(length, breadth):

    return length * breadth


print("Area:", area_rectangle(10, 5))
```

```python
#7. Simple Interest (default rate = 5%)

def simple_interest(p, t, r=5):

    return (p * t * r) / 100


print("SI:", simple_interest(1000, 2))



#8. Power of a number (default exponent = 2)

def power(num, exp=2):

    return num ** exp


print(power(5))

print(power(5, 3))



#9. Sum of any number of values using *args

def sum_values(*args):

    return sum(args)


print(sum_values(10, 20, 30))
```

```python
#10. Largest value using *args
def largest_value(*args):
    return max(args)

print(largest_value(5, 12, 7, 25))



#11. Display all key-value pairs using **kwargs
def show_details(**kwargs):
    for key, value in kwargs.items():
        print(key, ":", value)

show_details(name="Ravi", age=20, city="Chennai")



#12. Display employee details using **kwargs
def employee_details(**kwargs):
    for key, value in kwargs.items():
        print(key, ":", value)

employee_details(id=101, name="Anita", salary=30000)



#13. Lambda to find square
square = lambda x: x * x
```

```
print(square(6))


#14. Lambda to find larger of two numbers

larger = lambda a, b: a if a > b else b

print(larger(10, 25))


#15. Convert Celsius to Fahrenheit using map()

celsius = [0, 20, 30, 40]

fahrenheit = list(map(lambda c: (c * 9/5) + 32, celsius))

print(fahrenheit)


#16. Squares of list elements using map()

nums = [1, 2, 3, 4]

squares = list(map(lambda x: x**2, nums))

print(squares)


#17. Extract even numbers using filter()

nums = [1, 2, 3, 4, 5, 6]

even_nums = list(filter(lambda x: x % 2 == 0, nums))

print(even_nums)
```

```
#18. Extract numbers greater than 100 using filter()
nums = [50, 120, 80, 200, 30]
greater = list(filter(lambda x: x > 100, nums))
print(greater)
```

```
#19. Recursive function to calculate power
def power(base, exp):
    if exp == 0:
        return 1
    return base * power(base, exp - 1)

print(power(2, 5))
```

```
#20. Recursive Fibonacci series up to N terms
def fibonacci(n):
    if n <= 1:
        return n
    return fibonacci(n-1) + fibonacci(n-2)

n = 7
for i in range(n):
    print(fibonacci(i), end=" ")
```